

# SOCIOL 723: Problem Set 2

**Assigned:** Friday, September 18 • **Due:** Friday, October 2, 11:59PM

Your name:

**Instructions.** Submit a single PDF rendered from a Quarto or RMarkdown source, together with the source file. This problem set is about what you will do in research: reading a published table, fitting the same kind of model on real data, and writing up what it says. No question asks for a derivation; every computation uses the functions from Labs 3 and 4. Where a question asks for sentences, write them as you would in a paper. You may use AI assistance, but you are responsible for every line you submit. Use `set.seed(723)` wherever there is randomness. The GSS extract is `Data/gss_earnings.rds`; the school data are `nlme::MathAchieve` and `nlme::MathAchSchool`, merged as in Lab 4.

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## Reading a Published Logit: Killewald (2016)

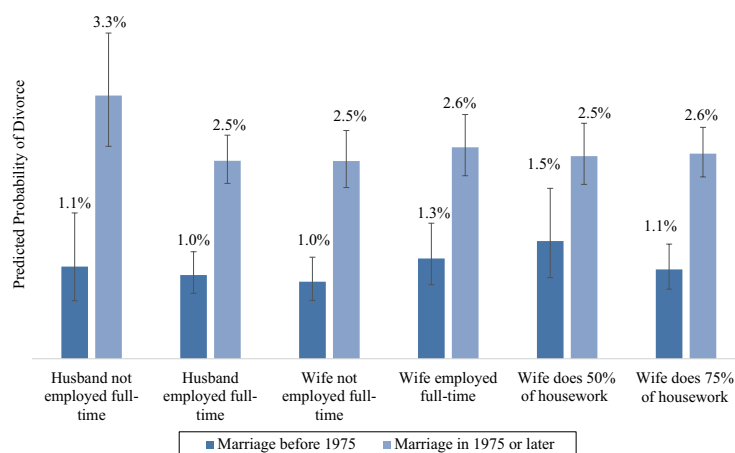
Does a marriage's economic situation predict whether it ends? Killewald weighs four answers sociologists have given: wives who could support themselves after a divorce are freer to leave; financial strain wears a marriage down; marriages are more stable when spouses follow traditional roles; or they are more stable when only one spouse works full-time, whoever that is. She follows more than 6,300 couples in the Panel Study of Income Dynamics from 1968 to 2013. One row of her data is one couple in one year, and the outcome is 1 in the year a couple divorces and 0 in every earlier year. Because the meaning of a wife's paycheck has changed, she fits the model separately to marriages begun in 1974 or earlier and in 1975 or later. Table 3 reports the logits; Figure 1 turns the key coefficients into predicted probabilities of divorce in the next year. Her headline finding: in recent marriages a husband who is not employed full-time raises the risk of divorce, while the wife's employment and share of housework no longer matter.

*Reference:* Killewald, Alexandra. 2016. "Money, Work, and Marital Stability: Assessing Change in the Gendered Determinants of Divorce." *American Sociological Review* 81(4): 696–719. The excerpt of Table 3 and Figure 1 are reproduced on the next page; the full table has about forty rows of controls.

1. **One coefficient, three scales.** Take the husband's full-time employment in the later cohort,  $-.293$ . Write one sentence on each scale: log-odds, odds (use the printed odds ratio), and probability (from Figure 1: 2.5% when he is employed full-time, 3.3% when he is not). Then compute the odds ratio implied by those two probabilities and compare it with the printed .746. Should the two agree exactly? Answer in one or two sentences, using the note under Figure 1 on how the probabilities were computed.

|                                | Married 1974 or earlier |        |       | Married 1975 or later |        |       |
|--------------------------------|-------------------------|--------|-------|-----------------------|--------|-------|
|                                | $\hat{\beta}$           | (SE)   | OR    | $\hat{\beta}$         | (SE)   | OR    |
| Wife employed full-time        | .267                    | (.153) | 1.306 | .069                  | (.088) | 1.071 |
| Husband employed full-time     | -.097                   | (.208) | .908  | -.293**               | (.106) | .746  |
| Wife's proportion of housework | -1.117*                 | (.463) | .327  | .050                  | (.233) | 1.051 |
| Constant                       | -7.476***               | (.884) |       | -3.525***             | (.393) |       |
| Couple-years                   | 32,853                  |        |       | 33,470                |        |       |

Killewald (2016), Table 3, excerpt. Logit coefficients, standard errors, and odds ratios as printed. \* $p < .05$ ; \*\* $p < .01$ ; \*\*\* $p < .001$ . Controls include marital duration, religion, race, home ownership, children, both spouses' education, region, and year of marriage.



**Figure 1.** Predicted Probabilities of Divorce in the Next Year  
*Note:* Probabilities are predicted based on the logistic regression coefficients in Table 3 for a respondent with all other covariates set to the cohort-specific means. Error bars are 95% confidence intervals, estimated by calculating the bounds of the confidence interval for the predicted log odds (non-bootstrapped) and then transforming those bounds to predicted probabilities.

Killewald (2016), Figure 1, reproduced for classroom use.

2. **A coefficient that is not significant.** In the later cohort, the wife's full-time employment has coefficient .069 and standard error .088; Figure 1 shows 2.6% when she is employed full-time and 2.5% when she is not. Write the sentence that belongs in a results section. In the earlier cohort the same coefficient is .267 with standard error .153, and Figure 1 shows 1.3% against 1.0%. Killewald writes that the association declined across cohorts but that the decline is not statistically significant. Compute the difference between the two coefficients and its standard error,  $\sqrt{.153^2 + .088^2}$ , and use them to explain in two sentences what supports each half of her claim.
3. **Units matter.** The wife's proportion of housework runs from 0 to 1, and its coefficient in the earlier cohort is -1.117, printed odds ratio .327. What change in housework does that odds ratio describe? Figure 1 plots a smaller change, from 50% to 75%. Start from the

figure's 1.5% at 50%, multiply the odds by  $e^{-1.117 \times .25}$ , and convert back to a probability; you should land on the figure's 1.1%. Then one sentence on why you cannot recover the odds ratio precisely by working backwards from the printed 1.5% and 1.1%, and one sentence on why a coefficient's size means nothing until you know the units of its variable.

## Your Own Logit, Reported Like a Paper

Fit the logit from lecture on the GSS extract:

```
m1 <- glm(ba ~ pareduc + wordsum + female + black, data = gss, family = binomial)
```

1. **Table and figure.** Report the coefficient table with odds ratios added, laid out like Killewald's Table 3. Then make your own Figure 1: with `avg_predictions`, compute the average predicted probability of a degree when everyone's parents have 8, 12, 16, and 20 years of education, with 95% intervals, and plot the four bars with error bars. Two sentences: why is the gain from 12 to 16 years not the same as the gain from 16 to 20, when the coefficient is one number?
2. **An interaction, read correctly.** Does parents' education matter more for women? Fit `ba ~ pareduc * female + wordsum + black`. Report the product coefficient with its  $z$ , then the AME of `pareduc` separately for men and women and the difference with its standard error. Answer the question in a short paragraph and say why the product coefficient alone could not have answered it.
3. **A block of controls.** Add the region dummies (`region`) to `m1`. Report the likelihood-ratio test, AIC, and BIC for the two models. In a short paragraph, say what each is measuring, why with  $n = 3,509$  the LR test rejects almost any block, and which model you would report and why.
4. **The results paragraph.** Write the paragraph (120 to 180 words) that would appear in the Results section of a paper reporting this model. Use predicted probabilities or AMEs, not odds ratios; give the reader magnitudes and uncertainty; do not mention the log-odds scale.

## Counts

Fit the Poisson regression for the number of children:

```
m_pois <- glm(childs ~ educ + female + black + exper + evermar, data = gss, family = poisson)
```

1. **Reading it, and which standard errors.** Interpret  $e^{\hat{\beta}}$  for `educ` and for `evermar`, one sentence each, in the form you would use in a paper. Then run `AER::dispersiontest(m_pois)` and report the standard error of `evermar` from the Poisson fit, from a quasi-Poisson fit, and from `sandwich::vcovHC`. Two sentences: what does the test say about the Poisson's

variance assumption, and which standard error would you report as your default and why?

2. **Zeros, and a model with two parts.** Report the observed share of childless respondents and the share the Poisson predicts (average `dpois(0, fitted(m_pois))`). Fit the hurdle model with `pscl::hurdle(..., dist = "poisson")` and the same covariates in both parts. Make one table with the log-likelihood, number of parameters, AIC, and BIC of the Poisson and the hurdle. Then, in a paragraph written for a reader, say what **evermar** does to the chance of *any* child, what it does to the number of children *among parents*, and what the single Poisson coefficient was averaging over.

## Observations in Groups

Merge `nlme::MathAchieve` with `nlme::MathAchSchool` as in Lab 4, and convert `School` to a plain factor with `factor(as.character(School))`, as the lab does: 7,185 students in 160 schools, math achievement, student SES, and school sector.

1. **From clustered standard errors to a random intercept.** Fit `lm(MathAch ~ SES + Sector)` and report the coefficient on `Sector` with its plain standard error and with its standard error clustered on `School`. In a short paragraph, explain to a first-year student why the plain standard error is wrong for this variable: sector is a property of the school, so it varies across 160 schools, not across 7,185 students. Then fit `lmer(MathAch ~ SES + Sector + (1 | School), REML = FALSE)` and report the two coefficients with standard errors,  $\hat{\tau}^2$ , and  $\hat{\sigma}^2$ . Say in one sentence each what the two variances measure, compute the intraclass correlation, and close with two sentences on what the random-intercept model gives you that the clustered standard error did not.
2. **Three estimates of one slope, and three questions.** Fit the fixed-effects regression `lm(MathAch ~ School + SES + Sector)`, with the school dummies listed *before* `Sector`. One coefficient comes back NA: which one, and why? What would you have concluded had you listed `Sector` before `School` and read off the number R printed? Then compare the SES slope under pooled OLS, the random intercept, and fixed effects in one three-column table, and explain in a paragraph where the random-intercept slope sits relative to the other two and why, in terms of the variation each estimate uses. Finally, for each research question below, say in two sentences which of *OLS with clustered standard errors*, *fixed effects*, or a *random intercept* you would use, naming the feature of the question that decides it.
  - (a) How much of earnings inequality in a country is between workplaces rather than within them, and is that share rising?
  - (b) Does a worker's wage rise after their own firm signs a union contract, using several years of data on the same workers and firms?
  - (c) Do students learn more in Catholic schools, using these 160 schools, where sector is a

property of the school?